

SOFTWARE REQUIREMENTS SPECIFICATION

Nagar Palika Online Recruitment Portal

Patan Municipal Council | Palanpur Municipal Council

Version 1.0 — Draft

Document Title	SRS — Nagar Palika Online Recruitment Portal
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Prepared For	Patan Nagar Palika & Palanpur Nagar Palika
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Deployment Model	Shared server, two subdomains (one per municipality)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for the Nagar Palika Online Recruitment Portal, a web-based government recruitment system to be deployed for Patan Municipal Council and Palanpur Municipal Council. The document is intended for use by developers, system architects, project managers, and municipal stakeholders.

1.2 Scope

The portal enables citizens to register once using their Aadhaar card, browse open job advertisements published by either municipality, submit applications, pay fees, and download call letters — all online. Municipal administrators manage advertisements, review applications, publish results, and upload notifications through a secure backend panel.

Key scope boundaries:

- Two independent subdomains on a shared server: `patan.domain.gov.in` and `palanpur.domain.gov.in`
- Each subdomain serves one municipality with its own data, admin panel, and branding

- One Aadhaar card = one registration (enforced system-wide per subdomain)
- Online fee payment via integrated payment gateway
- Bilingual interface: English and Gujarati

1.3 Definitions and Abbreviations

Term / Abbreviation	Definition
OTR	One Time Registration — a unique applicant profile linked to one Aadhaar card
Aadhaar	12-digit national identity number issued by UIDAI, used as primary identifier
Advt No	Advertisement Number — unique identifier for each job posting
Call Letter	Admit card issued to eligible candidates after fee verification
Admin	Authorised municipal officer with backend access
Department	Organisational unit within the municipality posting a vacancy
Captcha	Challenge-response test to prevent automated form submissions
SRS	Software Requirements Specification (this document)

1.4 References

- OJAS — Online Job Application System (ojas.gujarat.gov.in) — reference UX model
- SSC Online Portal (ssc.nic.in) — reference for exam/recruitment workflow
- Aadhaar Authentication API — UIDAI
- Payment Gateway Integration Guide (provider TBD)

2. Overall System Description

2.1 Product Perspective

The portal is a standalone web application hosted on a Linux server. Two subdomains point to the same codebase, but each subdomain operates on a logically isolated data partition (separate databases or schema-per-tenant). The system replaces manual or offline application processes currently used by both Nagar Palikas.

2.2 System Architecture Overview

- Server: Single physical/virtual server purchased by the municipality
- Subdomains: patan.domain.gov.in and palanpur.domain.gov.in
- Backend: Web application server with REST API layer
- Database: Separate database schemas per municipality for data isolation
- Storage: Document and image storage (photos, signatures, PDFs)
- Payment: Third-party payment gateway integration
- OTP & Notification Delivery: WhatsApp (primary, cost-optimised) + SMS fallback + Email

2.3 User Roles

Role	Description
Public / Guest	Unauthenticated visitor — can browse notices, quick links, help
Registered User	Citizen with a completed OTR profile; can apply, pay, download call letters
Administrator	Municipal officer — manages advertisements, applications, notifications, reports

2.4 Operating Constraints

- All public-facing pages must be accessible without JavaScript for basic information
- Portal must comply with Government of India accessibility guidelines (GIGW)
- Aadhaar integration must comply with UIDAI regulations and data protection norms
- Session timeout: 30 minutes of inactivity for authenticated users

3. Functional Requirements

3.1 Home Page

3.1.1 Navigation Bar 1 — Branding

The topmost bar displays municipal identity elements:

- Municipality logo (Ashoka emblem / Nagar Palika seal)
- Official name of the municipality in Gujarati and English
- Government of Gujarat emblem (as applicable)

3.1.2 Navigation Bar 2 — Main Menu

A persistent navigation bar containing the following links:

Menu Item	Behaviour / Sub-items
Home	Returns user to the home page
Registration	Dropdown with three options: Apply (new OTR), Edit (modify existing), Find (retrieve registration number)
Online Application	Dropdown with three options: Apply (submit application), Edit (modify draft), Print (print submitted application)
Fee	Opens the fee payment page
Call Letter	Allows eligible candidates to download call letters
Help	Opens the Help / Query page with contact info and FAQ
Login / Register	Opens the authentication modal (Login or New Registration)

3.1.3 Header Section

- Main banner image relevant to the municipality
- Scrolling ticker bar 1: Toll-free helpline contact number

- Scrolling ticker bar 2: One Time Registration (OTR) status notice (e.g., 'OTR Registration is Live')

3.1.4 Notice Board

A section listing recent notifications published by the administrator. Each notice entry includes:

- Notice title
- Date of publication
- Link to view or download the notice (PDF)

3.1.5 Important Instructions

A static section containing key instructions for applicants, editable by the administrator from the backend panel.

3.1.6 Quick Links

A grid of quick-access links:

- How to Register
- How to Apply
- Apply (direct link to application)
- Admit Card / Call Letter
- Results
- Help / Query

3.2 Registration Module

3.2.1 Overview

The Registration module implements One Time Registration (OTR). A citizen may register only once per subdomain using a valid Aadhaar card. On successful registration, the system generates a unique Registration ID. This ID is linked permanently to the Aadhaar number; duplicate registrations using the same Aadhaar are rejected.

3.2.2 Registration — Apply (New Registration)

Step 1: Understanding Registration

- Full-page display of instructions in Gujarati and English (Insight & General Instructions / સમજ અને જાહેર સૂચન)
- User must scroll to the bottom and click 'I Agree / હું સ્વીકાર કરું છું' to proceed

Step 2: Aadhaar Verification

- User enters 12-digit Aadhaar number
- OTP is sent to the mobile number registered with UIDAI
- User enters OTP; system verifies via UIDAI API
- On success, Aadhaar-linked basic data is pre-filled where available

- Phone number verification (independent of Aadhaar): user enters their personal mobile number; a 6-digit OTP is delivered via WhatsApp (primary) or SMS (fallback); user enters OTP to confirm ownership of the number before proceeding
- OTP delivery channel priority: WhatsApp message (zero marginal cost on approved templates) → SMS (fallback if WhatsApp undelivered after 30 seconds); all future notifications to this number also follow the same channel priority
- User creates a password for future login (min. 8 characters, one uppercase, one digit, one special character)

Step 3: Personal Details (ઉમેદવારની માહિતી)

- Full name (as per Aadhaar)
- Father's / Husband's name
- Date of birth
- Gender
- Category (General / OBC / SC / ST / EWS)
- Nationality
- Religion

Step 4: Communication Details (સંદેશ-વ્યવહારની માહિતી)

- Permanent address (house number, street, locality, taluka, district, state, PIN)
- Current / correspondence address (or checkbox: same as permanent)
- Mobile number (OTP-verified)
- Alternate mobile number (optional)
- Email address (OTP-verified)

Step 5: Other Details (અન્ય માહિતી)

- Marital status
- PH (Persons with Disability) status — if yes, type and percentage
- Ex-serviceman status
- Highest educational qualification

Step 6: Language Details (ભાષાની માહિતી)

- Languages known: Read / Write / Speak checkboxes for each
- Mother tongue

Step 7: Upload Photograph (ફોટો અપલોડ કરો)

- File format: JPG / JPEG
- Max size: 50 KB
- Dimensions: 3.5 cm × 4.5 cm (passport size)
- Background: white / light-coloured

Step 8: Upload Signature (સહી અપલોડ કરો)

- File format: JPG / JPEG

- Max size: 20 KB
- Dimensions: 3.5 cm × 1.5 cm

Step 9: Declaration

- Applicant reads and accepts the declaration statement
- Checkbox: 'I hereby declare that all information provided is true and correct'

Step 10: Captcha & Save

- Image-based captcha presented to the user
- On successful submission: system generates a unique Registration ID
- Registration ID displayed on screen and sent via SMS and email
- One Aadhaar = One Registration ID (system rejects duplicate Aadhaar on subsequent attempts)

3.2.3 Registration — Edit

- Accessible with Registration ID + Date of Birth, or Aadhaar + OTP
- Editable window: 48 hours from initial registration (exact window to be confirmed by municipality)
- Fields editable: communication details, photograph, signature, language details
- Non-editable after save: Aadhaar number, date of birth, name (as per Aadhaar)
- Any edit triggers an OTP confirmation before saving

3.2.4 Registration — Find

- User can retrieve Registration ID by entering: Mobile Number + Date of Birth, or Aadhaar Number + Date of Birth
- Registration ID sent to registered mobile and email after OTP verification

3.3 Online Application Module

3.3.1 Apply — New Application

Flow:

1. User lands on 'Enter Registration Number' page
2. User enters Registration ID + Date of Birth
3. Option A — Apply with OTR: pre-fills personal details from OTR profile, user proceeds to apply
4. Option B — Skip: user manually fills required fields without OTR pre-fill
5. User selects department from dropdown ('Select Advertisement by Department')
6. System lists all active/open posts under that department in a table showing: Advt No, Post Title, Last Date (dd/mm/yyyy hh:mm), Fee, Contact Info
7. User clicks 'Details' to view a detailed PDF of the post
8. User clicks 'Apply' against desired post
9. Application form is presented — currently contains a Name field as placeholder; full field list to be defined in the next revision (see open question in Section 9)
10. User clicks 'Apply Now' to submit
11. On successful submission: Application Reference Number is generated and displayed

3.3.2 Apply — Edit Application

- Editable only before the advertisement closing date
- User can modify preference order, update contact details
- Changes are logged with timestamp

3.3.3 Apply — Print Application

- User can generate a printable PDF of their submitted application
- PDF includes: Registration ID, Advt No, post name, personal details, photo, signature, application reference number

3.4 Fee Payment Module

- User enters Aadhaar number or Registration ID
- System retrieves list of all posts applied for under this registration
- For each post: post name, Advt No, fee amount, payment status (Paid / Pending)
- User selects posts for which payment is pending and clicks 'Pay Fee'
- Redirect to payment gateway (payment modes: UPI, Net Banking, Debit/Credit Card)
- On successful payment: fee receipt generated; status updated to Paid
- Fee receipt downloadable as PDF
- Failed payments: system retains application; user can retry payment

3.5 Call Letter / Admit Card Module

- User enters Registration ID + Date of Birth
- System checks eligibility (fee paid, application submitted, call letter published by admin)
- If eligible: call letter PDF displayed and downloadable
- Call letter includes: candidate name, registration ID, roll number, exam date, time, venue, instructions

3.6 Help / Query Page

- Toll-free helpline number prominently displayed
- Email contact for queries
- FAQ section (collapsible accordions)
- Step-by-step guides: How to Register, How to Apply
- Query submission form: name, registration ID, query category, description, submit

3.7 Administrator Panel

3.7.1 Authentication

- Separate admin login URL (not publicly indexed)
- Username + password + OTP (two-factor)
- Role-based access: Super Admin (full access), Department Admin (own department only)

3.7.2 Advertisement Management

Admin fills a structured form to publish a new post:

Field	Notes
Advt No (auto-generated)	Format: ORGCODE/YEAR/SEQ e.g. PATAN/202627/1
Post Title	In English and Gujarati
Department	Dropdown from configured departments
Class	Class I / II / III / IV
Pay Scale	e.g. 26000/-
Number of Vacancies	Total and category-wise breakup
Age Limit	Min–Max, or 'As per advertisement'
Educational Qualification	Text or 'As per detailed advertisement'
PH Description	Categories eligible, e.g. LV, HH, OL
Experience Required	Text or 'Not Applicable'
Application Fee	Amount in INR
Application Start Date	dd/mm/yyyy hh:mm
Application End Date	dd/mm/yyyy hh:mm
Probation Period	e.g. 5 years fixed pay
Detailed PDF Upload	Full advertisement document
Other Conditions	Free text or 'As per advertisement'
Status	Draft / Published / Closed / Archived

3.7.3 Notification Management

- Admin can publish/unpublish notices on the Notice Board
- Each notice: title, body text, optional PDF attachment, publish date, expiry date
- Admin can update Important Instructions section

3.7.4 Registration Management

- View all registered candidates (searchable and filterable by name, registration ID, date, category)
- View full profile of any candidate
- Export registrations to CSV / Excel

3.7.5 Application Management

- View all applications per advertisement
- Filter by department, post, category, fee status
- Download application list in CSV / Excel / PDF
- View individual application with candidate details

3.7.6 Fee Management

- View fee payment status per application

- Manual fee verification option (for offline/DD payments if applicable)
- Payment reconciliation report

3.7.7 Call Letter Management

- Upload roll number list (CSV) per advertisement
- Set call letter availability date and time
- Upload call letter template PDF or generate from template
- Enable / disable download per advertisement

3.7.8 Bulk Applicant PDF Export (ZIP Download)

After the application window for an advertisement closes, the administrator can generate and download a ZIP archive containing one PDF per applicant. Each PDF consolidates the complete candidate record for offline review, record-keeping, or forwarding to a selection committee.

- Trigger: Admin selects an advertisement and clicks 'Download All Applications (ZIP)'
- System generates one PDF per applicant; each PDF contains: Application Reference No, Advt No, Registration ID, full personal details, communication details, other details, language details, uploaded photograph, uploaded signature, list of applied posts and fee payment status, and declaration statement
- All individual PDFs are packaged into a single ZIP file named: Advtno_Applications_DDMMYYYY.zip (e.g. PATAN-202627-1_Applications_15062026.zip)
- Individual PDF filename format: RegID_ApplicantName_AdvtNo.pdf (e.g. 10001234_Rahul_Patel_PATAN-202627-1.pdf)
- Admin can filter before export: All applicants / Fee Paid only / Category-wise / Department-wise
- Large exports (500+ applicants) are generated asynchronously; admin receives a WhatsApp / email notification with a download link once the ZIP is ready
- Download link is available for 7 days; admin can regenerate at any time

4. Non-Functional Requirements

4.1 Performance

- Home page load time: < 3 seconds on a standard broadband connection
- Application submission response: < 5 seconds
- Concurrent users supported: minimum 500 simultaneous sessions per subdomain

4.2 Security

- All data in transit encrypted via HTTPS / TLS 1.2+
- Passwords stored using bcrypt or Argon2 hashing (never plaintext)
- SQL injection, XSS, and CSRF protections implemented
- Aadhaar data handled per UIDAI guidelines — not stored in raw form
- Session tokens expire on logout and after 30 minutes of inactivity
- Admin panel access restricted by IP whitelist (configurable)

4.3 Reliability & Availability

- Target uptime: 99.5% excluding scheduled maintenance

- Automated daily database backups with 30-day retention
- Maintenance window communicated 48 hours in advance via notice board

4.4 Usability

- Bilingual interface: Gujarati (primary) and English
- Mobile-responsive design (works on smartphones and tablets)
- WCAG 2.1 Level AA accessibility compliance
- Consistent UI patterns modelled on OJAS / SSC for citizen familiarity

4.5 Scalability

- Codebase supports adding additional municipality subdomains with configuration only
- Database schema supports multi-tenant isolation

5. System Constraints

- Must operate on the single server purchased by the municipalities
- Domain and SSL certificate provisioned by the municipality's IT team
- Payment gateway vendor to be selected and contracted by the municipality
- Aadhaar-based OTP requires UIDAI AUA/KUA empanelment (municipality to arrange)
- Notification delivery via WhatsApp Business API (primary channel, cost-optimised), SMS (fallback), and Email (SMTP); system must support all three channels with configurable priority

6. Key Data Entities

Entity	Key Attributes
Candidate (OTR)	Registration ID, Aadhaar hash, name, DOB, gender, category, contact, photo, signature
Advertisement	Advt No, post title, department, vacancies, fee, dates, PDF, status
Application	Application Ref No, Registration ID, Advt No, submission timestamp, status
Fee Payment	Payment ID, Application Ref No, amount, gateway transaction ID, status, timestamp
Call Letter	Registration ID, Advt No, roll number, exam details, download timestamp
Notification	Notice ID, title, body, PDF, publish date, expiry, status
Admin User	User ID, name, role, department, credentials (hashed), last login

7. Assumptions and Dependencies

- Municipalities will provide official logos, seal images, and branding guidelines before development begins
- Aadhaar OTP verification will be handled via UIDAI-authorized API (AUA empanelment required)
- A payment gateway agreement will be signed before the fee module goes live
- SMS and email delivery services will be contracted; API credentials provided to the development team

- Municipality IT staff will manage server provisioning, DNS, and SSL configuration
- The portal is for recruitment and advertising purposes only — it does not conduct online examinations

8. Open Questions for Stakeholder Clarification

#	Question	Impact
1	What is the exact edit window for a completed registration? 48 hours or longer?	Registration module edit logic
2	Will fee payment be online only, or will offline (DD/challan) payments also be supported?	Fee module scope
3	Which Aadhaar integration method will be used — OTP-based or offline XML?	Registration flow and UIDAI compliance
4	Should both subdomains share admin credentials or have completely separate admin accounts?	Admin architecture
5	Is a separate Results publication module required (beyond notice board)?	Scope clarification
6	Who provides the detailed advertisement PDFs — admin uploads or system-generated?	Advertisement management
7	What payment gateway is preferred (Razorpay, PayGov, Paytm, or state portal)?	Fee module integration
8	What fields should the online application form capture (beyond the OTR profile)? e.g. exam centre preference, category declaration, qualification details, experience details. TBD by municipality.	Application form design — blocks dev of apply module
9	Will the municipality register a WhatsApp Business API account (Meta-approved BSP)? This is required to send OTPs and notifications via WhatsApp at scale.	WhatsApp OTP & notification module; determines fallback SMS cost

Appendix A: Registration Flow Summary

The following outlines the complete user journey through the OTR registration process:

Step	Screen / Page	Key Action
1	Instructions Page (સમજ અને જાહેર સૂચના)	Read instructions → Click 'I Agree'
2	Aadhaar Verification	Enter Aadhaar + OTP → Set password
3	Personal Details (ઉમેદવારની માહિતી)	Fill name, DOB, gender, category
4	Communication Details (સંદેશ-વ્યવહારની માહિતી)	Fill address, mobile, email
5	Other Details (અન્ય માહિતી)	PH status, qualification, ex-serviceman
6	Language Details (ભાષાની માહિતી)	Select languages and proficiency
7	Upload Photograph (ફોટો અપલોડ કરો)	Upload passport photo (JPG, max 50 KB)

Step	Screen / Page	Key Action
8	Upload Signature (સહી અપલોડ કરો)	Upload signature image (JPG, max 20 KB)
9	Declaration	Read and accept declaration statement
10	Captcha + Save	Solve captcha → Submit → Registration ID generated

Appendix B: Application Submission Flow Summary

Step	Screen / Page	Key Action
1	Enter Registration No	Enter Reg ID + DOB
2	OTR Verification	Apply with OTR (pre-fill) or Skip
3	Select Department	Choose department from dropdown
4	Post Listing	View open posts, click Details or Apply
5	Application Form	Confirm personal details, fill extras <i>Note: The application form fields are to be defined. For the current version, the form contains a single Name field as a placeholder. Additional fields (qualifications, preferences, category, exam centre choice, etc.) will be specified and added in a subsequent revision of this document.</i>
6	Apply Now	Submit → Application Reference No generated
7	Fee Payment	Pay via gateway → Receipt generated
8	Print Application	Download PDF of submitted application

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